

PERSONALIZED PROGRAMMING EDUCATION PLATFORM INTEGRATING ADAPTIVE METHODS AND GAMIFICATION

By

K.A.M.P.Kumarapeli

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Bachelor of Software Engineering Honours

Department of Electrical and Computer Engineering

Faculty of Engineering Technology

The Open University of Sri Lanka

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K.A.M.P.Kumarapeli

121444372

Supervisor: Dr. (Ms.) G.S.N. Meedin

Department of Electrical and Computer Engineering

Faculty of Engineering Technology

Date:14.07.2024

Declaration

I hereby declare that this research proposal is my original work and does not include, without proper acknowledgment, any material previously published or submitted for a degree or diploma at any other university or institution of higher learning. To the best of my knowledge and belief, it does not contain any material written or published by another person except where such acknowledgement is made within the text.

Signature: Mahisha Pabasara

Date: 14.07.2024

K.A.M.P.Kumarapeli

I have reviewed the proposal and confirm that it adheres to the provided proposal outline.
I hereby approve the proposal for submission.



Signature of the supervisor:

Date: 15/07/2024

G. S. N. Meedin

ABSTRACT

The traditional approach to programming education often employs a standardized method that fails to cater to the diverse learning styles and skill levels of students. This lack of personalization can result in disengagement and diminished interest in programming. This research aims to develop a personalized programming education platform that integrates adaptive learning methods and gamification. Adaptive learning customizes educational content to meet the individual needs of learners, thereby enhancing the effectiveness of the learning experience. Gamification incorporates game design elements to increase student engagement and motivation. By combining these strategies, the proposed platform seeks to improve learning outcomes and user satisfaction compared to traditional methods. This study will investigate the limitations of current platforms, identify learner needs, and propose a comprehensive solution to enhance programming education.

In this research, will conduct a thorough review of existing literature on adaptive learning and gamification, utilizing sources such as academic journals, conference papers, and book resources. Key databases include IEEE Xplore, Google Scholar, and SpringerLink. The literature review will focus on adaptive learning techniques, gamification strategies, and case studies of their application in programming education.

Data collection will involve gathering secondary data from existing studies, reports, and databases to understand the effectiveness of these methods. The proposed platform will be designed to adaptively respond to individual learner's performance and incorporate gamified elements to make learning more engaging. The goal is to create a programming education platform that is both personalized and motivating, leading to better learning outcomes and increased student engagement.

Keywords: Personalized Learning, Adaptive Learning, Gamification, Programming Education

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01. INTRODUCTION

In the fast-changing world of computer science education, there is a growing need for more effective and engaging ways to teach programming. Traditional programming learning platforms often use a standardized approach that does not consider the different learning styles and skill levels of each student (Jando et al., 2017),(Alhawiti and Abdelhamid, 2017),(Muhammad and Ariatmanto, 2021). This one-size-fits-all method can make learning difficult and uninteresting for many students, leading to a decline in the popularity of these platforms (Chrysafiadi et al., 2020). To improve this situation, new methods that better meet the needs of individual learners should be explored. By doing so, a more inclusive and engaging learning environment for all students will be ensured.

One promising solution is to integrate adaptive learning methods and gamification into programming education platforms (Zaric, 2019). Adaptive learning methods tailor the educational content and activities to the specific needs and preferences of each learner (Laksitowening et al., 2016). By constantly adjusting the learning experience based on a student's performance, these systems provide personalized support that helps students learn more effectively (Drissi and Amirat, 2016). Gamification, which involves using game design elements like points, badges, and challenges in educational contexts, can significantly increase student engagement and motivation (Khaleel et al., 2020),(Universitas Sebelas Maret, Indonesia et al., 2022),(Cuervo-Cely et al., 2022). However, these strategies are not widely used together in current programming education platforms, which limits their potential benefits.

The growing diversity of students in their backgrounds, prior knowledge, and learning preferences requires a move from traditional teaching methods to a more personalized approach. Studies have shown that personalized learning experiences can significantly improve educational outcomes. For example, adaptive learning platforms that adjust content difficulty based on student performance can help bridge the gap for students with varying levels of understanding (Xue and Zeng, 2020). This approach not only enhances comprehension but also keeps students motivated by presenting them with appropriate challenges.

Also, the integration of gamification in education has been proven to enhance student motivation and participation. By incorporating elements such as leaderboards, achievements, and interactive tasks, gamified education platforms can make learning more enjoyable and easier (Khaleel et al., 2020). Research has demonstrated that gamification can lead to higher levels of student engagement, improved retention rates, and better overall performance in educational settings (Universitas Sebelas Maret, Indonesia et al., 2022),(Cuervo-Cely et al., 2022).

Despite these advantages, the combination of adaptive learning and gamification in programming education is still minimally used. Most existing platforms either focus on one of these strategies or fail to implement them effectively together. This research aims to fill this gap by developing a comprehensive programming education platform that leverages both adaptive learning methods and gamification to create a personalized and engaging learning experience. By addressing these challenges, programming education can be improved, making it more effective for diverse learners.

01.1 Background of the Project

Traditional programming learning platforms often provide the same content and exercises to all learners, regardless of their individual abilities and learning preferences. This one-size-fits-all approach can cause students to lose interest, either because they struggle to keep up or because they find the material too easy. As a result, these platforms often fail to keep learners engaged. Adaptive learning methods, however, tailor the learning experience to each student's performance and preferences, offering a more personalized approach (Alhawiti and Abdelhamid, 2017),(Muhammad and Ariatmanto, 2021).

Incorporating adaptive learning into programming education means that the platform continuously assesses the learner's progress and adapts the content accordingly. For instance, if a student has trouble with a concept, the system can offer extra resources or exercises to help them understand better. Conversely, if a student shows proficiency, the system can present more advanced topics to keep them challenged and engaged (Chrysafiadi et al., 2020). This dynamic adjustment ensures that each learner receives the right level of difficulty and support, making the learning process more efficient and enjoyable.

Gamification involves using game design elements in non-game contexts like education to boost engagement and motivation. In programming education, gamification can make learning more interactive and fun, leading to greater engagement with the material (Zaric, 2019). However, many platforms have not fully utilized the potential of gamification, lacking effective implementation strategies (Universitas Sebelas Maret, Indonesia et al., 2022). Research suggests that combining adaptive learning with gamification can create a strong combination that enhances both engagement and learning outcomes (Khaleel et al., 2020).

The goal of this project is to develop a programming education platform that integrates these two strategies, addressing the shortcomings of traditional methods and providing a more effective and engaging learning experience for students. By combining the strengths of adaptive learning and gamification, the proposed platform aims to meet the diverse needs of learners and improve their overall educational experience

01.2 Aim

This research aims to propose a suitable personalized programming education platform that integrates adaptive learning methods and gamification to create a more effective, engaging, and personalized learning experience. The goal is to address the limitations of traditional programming education platforms by tailoring educational content to individual learners' needs and incorporating game design elements to enhance student engagement and motivation.

01.3 Objectives

- 1) Identify the adaptive learning methods used in the existing programming teaching education platforms
- 2) Identify the needs and preferences of learners to incorporate in the education platform
- 3) Develop a personalized programming education platform integrating adaptive learning methods and gamification elements.
- 4) Evaluate the effectiveness of the developing platform in improving learning outcomes and user satisfaction.

02. PROBLEM IDENTIFICATION

Current programming education platforms often fail to engage and effectively educate students due to their one-size-fits-all approach. These platforms typically do not address the varying learning styles, skill levels, and motivational factors of individual learners, leading to disengagement and poor learning outcomes (Jando et al., 2017),(Muhammad and Ariatmanto, 2021),(Xue and Zeng, 2020). Additionally, the lack of gamification elements in many platforms results in a less engaging and interactive learning experience, which is crucial for maintaining student interest and motivation (Khaleel et al., 2020). This research aims to address these issues by proposing a personalized programming education platform that leverages adaptive learning and gamification to create a more effective and engaging learning environment.

02.1 Problem Statement

Traditional programming education platforms often fail to meet the diverse needs of learners, resulting in disengagement and suboptimal learning outcomes. There is a need for a platform that personalizes learning experiences and incorporates gamification to enhance engagement and effectiveness.

02.2 Research Questions

1. How can adaptive learning methods be effectively integrated into programming education platforms to address individual learner needs?
2. What gamification elements are most effective in enhancing student engagement and motivation in programming education?
3. How can a programming education platform be designed to combine adaptive learning and gamification effectively?

03. LITERATURE REVIEW

The field of programming education has seen various approaches and methodologies aimed at improving learner engagement and effectiveness. This literature survey reviews key studies and findings related to adaptive learning and gamification in education, highlighting their potential benefits and the gaps that this research aims to address.

The document types and quantities of the relevant research studies analyzed in this research are shown in the table 03:1 below.

Document Types	Quantity
Journal Article	07
Conference Paper	02
Book	01
Book Section	01
Total	11

Table 03:1 – relevant research studies

The source database of the selected articles in the table above is as follows:

IEEE Xplore (<http://ieeexplore.ieee.org>)

Google Scholar (<http://scholar.google.com>)

ResearchGate (<https://www.researchgate.net>)

Springer Link (<link.springer.com>)

03.1 Adaptive Learning Methods

Adaptive learning methods adjust the educational content and activities based on individual student performance and learning styles (Muhammad and Ariatmanto, 2021),(Laksitowening et al., 2016). Studies describe the design of a personalized e-learning platform that adapts to the unique needs of each learner, significantly improving the learning experience (Chrysafiadi et al., 2020),(Xue and Zeng, 2020). The Study discuss optimizing programming language learning through student modeling in adaptive environments, emphasizing the need for more widespread adoption of such methods (Alhawiti and Abdelhamid, 2017). Study found that enhancing e-learning systems to support learning style-based personalization can lead to more effective learning outcomes (Laksitowening et al., 2016). Similarly, this review also supports the idea that adaptive systems, which adjust based on a student's learning style, can significantly enhance the learning experience (Jando et al., 2017).

Despite these benefits, the integration of adaptive learning in programming education remains limited. Many existing platforms still rely on static content that does not adapt to the diverse needs of learners, leading to disengagement and suboptimal learning outcomes.

This research aims to fill this gap by developing a programming education platform that leverages adaptive learning to provide a personalized learning experience.

03.2 Gamification

Gamification involves incorporating game design elements, such as points, badges, leaderboards, and challenges, into non-game contexts to enhance engagement and motivation. In educational settings, gamification has been shown to increase student participation, motivation, and enjoyment, leading to better learning outcomes (Khaleel et al., 2020). For example, platforms like SoloLearn, CodeHS and CodeGym have implemented gamification elements to create more engaging and interactive learning experiences (Cuervo-Cely et al., 2022). Study has found that gamification can lead to higher levels of student engagement and motivation, which are critical for effective learning (Khaleel et al., 2020). However, the use of gamification in programming education is often superficial, lacking the depth and integration needed to truly enhance the learning experience. (Universitas Sebelas Maret, Indonesia et al., 2022) By combining gamification with adaptive learning, this research seeks to create a more immersive and effective educational platform.

03.3 Combining Adaptive Learning and Gamification

The synergy between adaptive learning and gamification has the potential to create powerful educational experiences. Adaptive learning ensures that content is tailored to individual learners' needs, while gamification elements enhance engagement and motivation. Research suggests that this combination can lead to improved educational outcomes by addressing both the cognitive and emotional aspects of learning (Zaric, 2019). However, there is a lack of comprehensive studies and practical implementations that integrate both strategies effectively in programming education.

By examining these studies, it becomes evident that while both adaptive learning and gamification have significant potential to improve programming education, their combined application is not yet fully realized. This gap in the research underscores the need for a comprehensive approach that integrates both strategies to create a personalized, engaging, and effective learning platform for programming education.

03.4 Existing Educational Platforms used in teaching Programming

The landscape of programming education has evolved significantly over the years, with numerous platforms emerging to address the diverse needs of learners. These platforms leverage various methodologies, including traditional approaches, adaptive learning, and

gamification. These studies review some of the leading programming education platforms currently in use, highlighting their features and educational strategies (Universitas Sebelas Maret, Indonesia et al., 2022),(Cuervo-Cely et al., 2022).

Table 3.2 provides information on some of the leading programming education platforms currently available and information about these education platforms is taken from studies (Universitas Sebelas Maret, Indonesia et al., 2022) and (Cuervo-Cely et al., 2022).

Name	Type	Programming Languages	Gamified Elements
SoloLearn	Mobile and Web	Python, C++, Java, JavaScript, C#, PHP, Swift, Ruby, jQuery C, HTML, CSS, SQL	Points, Trophies, Levels, Peer Competition, Social Interaction Areas, Leaderboards, and Badges
CheckiO	Web	Python and JavaScript	Levels, Challenges, Leaderboards, Social Interaction Areas, Points
Code Combat	Web	Python, JavaScript, HTML and CSS	Levels, Avatars, Virtual Worlds, Points, Challenges, Multiple Attempts, Leaderboards, Clans, Badges, Missions, Rewards, Progress, Coins, Customization, Social Area, and Rankings
CodeGym	Web	Java	Challenges, Game Restrictions, Ways to Earn or Lose Points, Characters, Narrative Scenarios, Unlimited Attempts, Badges, Teams, Performance Stats, Leaderboards, Social Interaction Areas, Levels, and Access to Locked Content
CodeHS	Web	Java, Javascript, HTML, Python, CSS, SQL, Karel, and React	Points, Badges, Progress Bars, and Levels

Table 03:2 - programming education platforms currently available

04. METHODOLOGY

The methodology section outlines the research design and procedures used to develop and evaluate the proposed personalized programming education platform. This platform aims to integrate adaptive learning methods and gamification to enhance student engagement and learning outcomes. The methodology consists of three main phases: data collection and study of existing educational platforms, platform development and system evaluation.

04.1 Data collection and study of existing educational platforms

The data collection process focused on collecting secondary data from existing studies, reports and databases. The following steps outline the process of collecting and using this data:

01. Identification of relevant studies and reports

- Search for and identify key studies, reports, and articles on adaptive learning, gamification, and programming education platforms.
- Use academic databases such as IEEE Xplore, Google Scholar, and SpringerLink to find relevant literature.
- Prioritize sources that provide comprehensive data on the effectiveness of adaptive learning and gamification in educational contexts.

02. Extraction of data

- Extract relevant data from identified studies and reports, focusing on metrics such as learner engagement, motivation, performance, and retention.

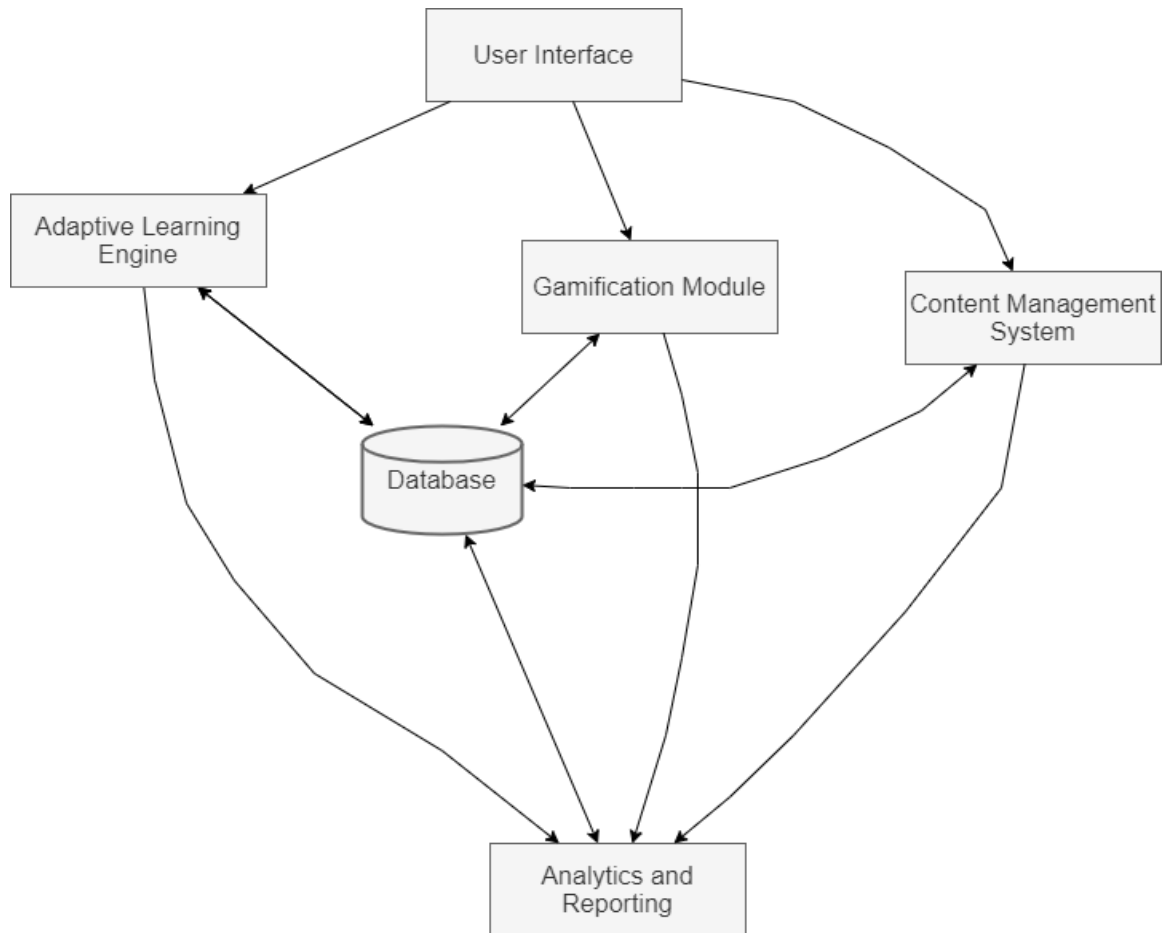
03. Data categorization

- Organize the extracted data into categories based on themes such as adaptive learning techniques, gamification strategies, and their impact on learning outcomes.

04.2 Platform development

A conceptual diagram illustrating the integration of adaptive learning methods and gamification within the platform is shown in Figure 04.1 below. This provides an overview of the core components of the proposed programming learning platform.

Figure 04:1 - conceptual diagram of Platform



1. User Interface: The front-end through which learners interact with the platform.
2. Adaptive Learning Engine: Customizes content based on individual learner's performance and preferences.
3. Gamification Module: Incorporates game elements such as points, badges, leaderboards, and challenges to enhance engagement.
4. Content Management System: Manages and delivers educational content.
5. Analytics and Reporting: Tracks learner progress and provides feedback for continuous improvement.
6. Database: Stores user data, learning materials, and performance metrics.

04.3 System evaluation

The following methods will be included in the evaluation of the platform.

01. Usability testing

- Conduct usability testing with a sample group of learners to assess the user-friendliness of the platform.
- Gather feedback on user interface, navigation and overall user experience.

02. Performance appraisal

- Measuring the impact of the platform on learner engagement, motivation and performance.
- Using pre- and post-tests, surveys and interviews to collect data on learning outcomes.

03. Comparative analysis

- Comparing the performance of learners using the platform with those using traditional programming education methods.
- Analyzing data to determine the effectiveness of adaptive learning and gamification integration.

04. Continuous improvement

- Using evaluation results to make continuous improvements to the platform.
- Updating the platform based on the latest research and technological advances in adaptive learning and gamification.

05. TIMELINE OF RESEARCH PROPOSAL

Table 05:1 shows the information about the schedule of the research.

Activity	February				March				April				May				June				July	
	1 w	2 w	3 w	4 w	1 w	2 w	3 w	4 w	1 w	2 w	3 w	4 w	1 w	2 w	3 w	4 w	1 w	2 w	3 w	4 w	1 w	2 w
Picking research topic.																						
Finding research gaps that relevant for picked topic.																						
Identify Research Problem.																						
Writing the research proposal																						
Finish writing the research proposal and get the supervisor's approval.																						

Table 05:1 - timeline of research proposal

06. CONCLUSION

This project aims to develop a personalized programming education platform that combines adaptive learning and gamification. Traditional methods often fail to address the different learning styles and skill levels of students, leading to disengagement and poor outcomes. By using adaptive learning, the platform customizes the educational content to fit each student's needs, ensuring they remain challenged and engaged. Gamification elements like points, badges, and leaderboards make learning fun and motivating.

The proposed platform addresses the limitations of traditional programming education by providing a more effective and enjoyable learning experience. It helps students learn better by adapting to their performance and making learning interactive and enjoyable. This approach is expected to improve learning efficiency, student satisfaction, and retention rates. Combining adaptive learning with gamification offers a promising solution to enhance programming education, making it more engaging and effective for diverse learners.

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